Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 2: Arithmetic Expressions

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	2 — Arithmetic Expressions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every item is a national-Olympiad multi-step trap: deeply nested brackets, full sign-flips on bracket removal, distributive-structure reasoning, and compare-without-computing puzzles whose wrong options are exactly the values produced by left-to-right slips, half-distribution, and unflipped signs. Do not write on this paper — mark answers on your answer sheet.

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1. Evaluate: $200 - 4 \times (15 - 2 \times (8 - 5)) + 6$.

(A) 170
(B) 158
(C) 242
(D) 164

2. Without computing either product, which is greater: 56×48 or $56 \times 50 - 56 \times 2$?

(A) They are equal
(B) 56×48
(C) $56 \times 50 - 56 \times 2$
(D) Cannot be decided

3. Which removal of brackets is CORRECT for $150 - (30 - 12 + 8 - 5)$?

(A) $150 - 30 - 12 + 8 - 5$
(B) $150 - 30 + 12 - 8 + 5$
(C) $150 - 30 + 12 + 8 - 5$
(D) $150 - 30 - 12 - 8 + 5$

4. Which expression equals $14 \times 100 - 14 \times 3$?

(A) 14×97
(B) $14 \times 97 - 3$
(C) 14×103
(D) $1400 - 3$

5. Evaluate: $120 \div (2 + 4 \times 2) \div 2 + 3 \times (11 - 2 \times 4)$.

(A) 15 (B) 9
(C) 12 (D) 33

6. Written as a sum of signed terms, the expression $90 - 4 \times 5 + 36 \div 9 - 7$ has which terms?

(A) 90, −4, 5, 36, 9, −7 (B) 90, −20, −4, −7
(C) 90, 20, 4, 7 (D) 90, −20, 4, −7

7. Without fully computing, which is greater: $73 - 28$ or $75 - 31$?

(A) $75 - 31$ (B) They are equal
(C) $73 - 28$ (D) Cannot be decided

8. Which single bracketed expression equals $100 - 24 - 16$?

(A) $100 - (24 - 16)$ (B) $(100 - 24) + 16$
(C) $100 - (24 + 16)$ (D) $100 + (24 - 16)$

9. Evaluate: $5 \times (20 - (8 + 2 \times 3)) - 4$.

(A) 26 (B) 30
(C) 34 (D) 46

10. Using the distributive property, $6 \times 48 + 6 \times 52$ equals:

(A) 100 (B) 300
(C) 600 (D) 6000

11. To evaluate $36 - 19 + 14 - 6$ by grouping the added terms and the subtracted terms, which grouping is valid and gives the value?

(A) $(36 - 19) + (14 - 6) = 25$ only by luck, but $(36 + 14) + (19 + 6) = 75$
(B) $(36 + 19) - (14 + 6) = 35$ (C) $(36 + 14) - (19 + 6) = 25$
(D) order is fixed, cannot regroup = 25

12. Sunita rewrites $80 - (25 - 10 - 5)$ without brackets. Which is correct?

(A) $80 - 25 - 10 - 5 = 40$ (B) $80 - 25 + 10 - 5 = 60$
(C) $80 - 25 + 10 + 5 = 70$ (D) $80 + 25 - 10 - 5 = 90$

13. Without computing either product fully, which is greater: 27×41 or 28×40?

(A) 27×41 (B) 28×40
(C) They are equal (D) Cannot be decided

14. Evaluate: $72 \div (3 + 3 \times 2) \times 4 - 2 \times (9 - 2 \times 3)$.

(A) 26 (B) 2
(C) 18 (D) 38

15. Using the distributive property to compute 23×101, the correct split and value are:

(A) $23 \times 100 + 1 = 2301$ (B) $23 \times 100 + 23 \times 100 = 4600$
(C) $23 \times 100 + 23 \times 1 = 2323$ (D) $23 \times 100 - 23 = 2277$

16. Four adult tickets cost ₹120 each and three child tickets cost ₹80 each; a ₹50 coupon is taken off the whole bill. Which expression gives the amount paid, in ₹?

(A) $(4 \times 120 + 3 \times 80 - 50)$ read as $4 + 120 \times 3 + 80 - 50$

(B) $(4 + 3) \times (120 + 80) - 50$

(C) $4 \times 120 + 3 \times 80 - 50 \times 7$

(D) $4 \times 120 + 3 \times 80 - 50$

17. Which statement about $100 - (40 - (10 - 3))$ is TRUE?

(A) It equals $100 - 40 - 10 - 3 = 47$

(B) It equals $100 - 40 + 10 - 3 = 67$

(C) It equals $100 - 40 + 10 + 3 = 73$

(D) It equals $100 - 40 - 10 + 3 = 53$

18. Without adding, which is greater: $248 + 176$ or $251 + 174$?

(A) $248 + 176$

(B) $251 + 174$

(C) They are equal

(D) Cannot be decided

19. Evaluate: $3 \times (4 + 2 \times (7 - 3)) - 5 \times (6 - 2 \times 2)$.

(A) 46

(B) 6

(C) 36

(D) 26

20. Which pair of expressions is EQUAL, illustrating distribution over a subtraction?

(A) $9 \times (20 - 4)$ and $9 \times 20 - 4$

(B) $(9 - 20) \times 4$ and $9 - 20 \times 4$

(C) $9 \times (20 - 4)$ and $9 \times 20 + 9 \times 4$

(D) $9 \times (20 - 4)$ and $9 \times 20 - 9 \times 4$

21. Evaluate $7 \times 8 - 5 \times 4 + 6 \times 3 - 2$ by first collapsing each product to a signed term:

(A) 34

(B) 92

(C) 56

(D) 52

22. Which single bracketed expression equals $60 - 8 - 9 - 5$?

(A) $60 - (8 - 9 - 5)$

(B) $60 - (8 + 9 + 5)$

(C) $60 - (8 + 9 - 5)$

(D) $(60 - 8) - (9 + 5) + 5$

23. Without computing, which is greater: 35×35 or 34×36?

(A) 34×36

(B) 35×35

(C) They are equal

(D) Cannot be decided

24. Evaluate: $144 \div (12 \div (8 - 6)) + 2 \times 3$.

(A) 30

(B) 12

(C) 24

(D) 78

25. A crate holds 6 bottles. A shopkeeper buys 9 crates, then 4 bottles break. Which expression gives the bottles still intact, and its value?

(A) $9 \times (6 - 4) = 18$

(B) $(9 - 4) \times 6 = 30$

(C) $9 + 6 - 4 = 11$

(D) $9 \times 6 - 4 = 50$

26. Which expression equals 8×93?

(A) $8 \times 100 - 7$ (B) $8 \times 100 - 8 \times 7$
(C) $8 \times 90 + 3$ (D) $8 \times 100 + 8 \times 7$

27. Remove the brackets correctly: $50 + (30 - 10) - (8 - 3)$.

(A) $50 + 30 - 10 - 8 - 3$ (B) $50 + 30 - 10 - 8 + 3$
(C) $50 - 30 + 10 - 8 + 3$ (D) $50 + 30 + 10 - 8 + 3$

28. Evaluate: $100 - 48 \div 6 + 5 \times 4 - 12 \div 4$.

(A) 69 (B) 125
(C) 115 (D) 109

29. Without computing, which is greater: 47×99 or $47 \times 100 - 47$?

(A) 47×99 (B) They are equal
(C) $47 \times 100 - 47$ (D) Cannot be decided

30. Evaluate: $4 \times ((30 - 2 \times (7 + 1)) \div 2) + 5$.

(A) 61 (B) 28
(C) 33 (D) 23

31. A worker earns ₹90 an hour and spends ₹40 an hour over the same 7-hour shift. Which expression gives the net amount kept, in ₹?

(A) $7 \times (90 - 40)$ (B) $7 \times 90 - 40$
(C) $7 \times 90 - 40 \times 1$ (D) $(7 \times 90 - 7 \times 40) \times 7$

32. By the distributive property, 18×35 can be correctly split as:

(A) $18 \times 30 + 5$ (B) $18 \times 30 + 18 \times 5$
(C) $18 \times 30 \times 5$ (D) 18×35 cannot be split

33. Which expansion of $90 - (15 + 6 - 4) + 5$ is CORRECT?

(A) $90 - 15 + 6 - 4 + 5$ (B) $90 - 15 - 6 - 4 + 5$
(C) $90 - 15 - 6 + 4 + 5$ (D) $90 - 15 + 6 + 4 + 5$

34. Which of these expressions has the GREATEST value?

(A) $6 + 2 \times 5 - 1$ (B) $(6 + 2) \times (5 - 1)$
(C) $6 \times 2 + 5 \times 1$ (D) $6 \times 2 - 5 + 1$

35. Without subtracting, which is greater: $502 - 318$ or $499 - 314$?

(A) $502 - 318$ (B) They are equal
(C) $499 - 314$ (D) Cannot be decided

36. Evaluate: $250 - 3 \times (40 - (12 + 2 \times 4)) - 8$.

(A) 134 (B) 182
(C) 190 (D) 242

37. Five buses each carry 32 students and 2 teachers. Which expression gives the total number of people, in all?

(A) $5 \times 32 + 2$ (B) $5 + 32 \times 2$
(C) $(5 + 32) \times 2$ (D) $5 \times (32 + 2)$

38. Which expression equals $15 \times 7 + 15 \times 13$?

(A) $15 \times 20 + 15$ (B) 15×91
(C) 15×20 (D) $15 + 7 + 13$

39. Which equality is TRUE?

(A) $120 - (30 + 25) = 120 - 30 + 25$ (B) $120 - (30 - 25) = 120 - 30 - 25$
(C) $120 - (30 + 25) = 120 + 30 - 25$ (D) $120 - (30 + 25) = 120 - 30 - 25$

40. To add $19 + 36 + 81 + 64$ quickly by pairing into round numbers, the best grouping and total are:

(A) $(19 + 36) + (81 + 64) = 200$, but not round(B) $(19 + 64) + (36 + 81) = 200$
(C) order is fixed, cannot regroup (D) $(19 + 81) + (36 + 64) = 200$

41. Evaluate: $96 \div 8 \times 3 - 2 \times (5 + 6 \div 3) + 4$.

(A) 26 (B) 18
(C) 54 (D) −6

42. Without fully computing, which expression is the GREATEST?

(A) $25 \times 42 - 25 \times 2$ (B) $25 \times 38 + 25 \times 2$
(C) 25×41 (D) $25 \times 39 + 25$

43. Using the distributive idea, the value of $17 \times 6 + 17 \times 4 - 17$ is:

(A) 170 (B) 153
(C) 187 (D) 136

44. Which removal of brackets is CORRECT for $200 - (50 - 30 + 20 - 10)$?

(A) $200 - 50 + 30 - 20 + 10$ (B) $200 - 50 - 30 + 20 - 10$
(C) $200 - 50 + 30 + 20 - 10$ (D) $200 + 50 - 30 + 20 - 10$

45. A tank holds 500 litres. Water is drained on three occasions of 80 litres each, then 150 litres is added back. Which expression gives the litres now in the tank?

(A) $500 - 3 + 80 + 150$ (B) $500 - 3 \times 80 + 150$
(C) $500 - 3 \times (80 + 150)$ (D) $(500 - 3) \times 80 + 150$

46. In the expression $120 - 36 \div 3 \times 2 + 7$, what single signed term comes from $36 \div 3 \times 2$?

(A) −6 (B) −12
(C) +24 (D) −24

47. Evaluate: $(84 + 36) \div (2 + 4 \times 2) - 3 \times (10 - 2 \times 4)$.

(A) 6 (B) 18
(C) −12 (D) 54

48. Without finishing the arithmetic, which is greater: $3 \times (20 + 4)$ or $3 \times 20 + 4$?

(A) $3 \times (20 + 4)$ (B) $3 \times 20 + 4$

(C) They are equal (D) Cannot be decided

49. A carton contains 5 red pens and 3 blue pens. Which expression gives the total pens in 12 such cartons, and equals which distributed form?

(A) $12 \times (5 + 3) = 12 \times 5 + 3$ (B) $12 \times 5 + 3 = 12 \times (5 + 3)$

(C) $12 \times 5 \times 3 = 12 \times (5 + 3)$ (D) $12 \times (5 + 3) = 12 \times 5 + 12 \times 3$

50. Evaluate: $64 - [30 - (12 - 4) - 6]$.

(A) 32 (B) 40

(C) 48 (D) 56

51. There are 84 pencils packed 7 to a pouch; then 5 pouches are given away. Which expression gives the pouches left, and its value?

(A) $84 \div (7 - 5) = 42$ (B) $(84 - 5) \div 7 = 11$ (rounded)

(C) $84 \div 7 \div 5 = 2$ (rounded) (D) $84 \div 7 - 5 = 7$

52. Written as signed terms, the expression $60 - 4 \times 3 + 18 \div 2 - 5$ has which terms?

(A) 60, −12, 9, −5 (B) 60, −4, 3, 18, 2, −5

(C) 60, −12, −9, −5 (D) 60, 12, 9, 5

53. Evaluate: $5 + 2 \times (3 \times (4 + 2) - (10 - 2 \times 3))$.

(A) 49 (B) 28

(C) 25 (D) 33

54. Without computing, which is greater: 64×98 or $64 \times 100 - 64 \times 3$?

(A) $64 \times 100 - 64 \times 3$ (B) 64×98

(C) They are equal (D) Cannot be decided

55. Using $36 = 40 - 4$, the distributive computation of 25×36 gives:

(A) $25 \times 40 - 4 = 996$ (B) $25 \times 40 + 25 \times 4 = 1100$

(C) $25 \times 40 - 25 \times 4 = 800$ (D) $25 \times 40 - 25 \times 4 = 900$

56. Where must brackets be placed so that $20 - 6 + 2$ equals 12?

(A) $20 - (6 + 2)$ (B) $(20 - 6) + 2$

(C) $20 - (6 - 2)$ (D) $(20 - 6 + 2)$

57. Each shirt is priced ₹250 but carries a ₹30 discount per shirt. Which expression gives the cost of 8 shirts, in ₹?

(A) $8 \times 250 - 30$ (B) $8 \times 250 - 30 \times 1$

(C) $8 \times (250 - 30)$ (D) $8 \times (250 - 30) \times 8$

58. Evaluate: $144 \div 12 + 7 \times 3 - 50 \div 5 - 2 \times 2$.

(A) 27
(B) 39
(C) 19
(D) 47

59. Which expression has the SAME value as $(40 + 20) \div 6 + 3 \times 4$?

(A) $60 \div 6 + 12$
(B) $40 + 20 \div 6 + 12$
(C) $(40 + 20) \div (6 + 3) \times 4$
(D) $60 \div (6 + 3 \times 4)$

60. Without computing, which expression equals $200 - 99$?

(A) $200 - 100 - 1$
(B) $200 + 100 - 1$
(C) $200 - 100 + 1$
(D) $200 - 90 - 9 - 1$

Solutions — Mathematics (Class 7), Chapter 2: Arithmetic Expressions — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	D	31	A	41	A	51	D
2	A	12	C	22	B	32	B	42	C	52	A
3	B	13	B	23	B	33	C	43	B	53	D
4	A	14	A	24	A	34	B	44	A	54	B
5	A	15	C	25	D	35	C	45	B	55	D
6	D	16	D	26	B	36	B	46	D	56	A
7	C	17	B	27	B	37	D	47	A	57	C
8	C	18	B	28	D	38	C	48	A	58	C
9	A	19	D	29	B	39	D	49	D	59	A
10	C	20	D	30	C	40	D	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Innermost first: 8 − 5 = 3, then 2 × 3 = 6, so 15 − 6 = 9. Now 4 × 9 = 36, and 200 − 36 + 6 = 170. Subtracting the trailing +6 instead of adding it gives 158. Adding the product 4 × 9 instead of subtracting it gives 200 + 36 + 6 = 242. Dropping the +6 entirely (treating it as already used) gives 200 − 36 = 164.

2. (A) By the distributive idea 56 × 50 − 56 × 2 = 56 × (50 − 2) = 56 × 48, so the two are identical. Believing 56 × 50 − 56 × 2 is bigger comes from seeing the larger first product 56 × 50 and forgetting the full second subtraction; believing 56 × 48 is bigger comes from over-subtracting 56 × 2 twice.

3. (B) A minus sign in front of the bracket flips every inner sign: −30 stays −30, +12 becomes... each sign reverses so −(30 − 12 + 8 − 5) = −30 + 12 − 8 + 5. The version 150 − 30 − 12 + 8 − 5 flips only some signs (failing to flip −12 and +8); 150 − 30 + 12 + 8 − 5 leaves +8 unflipped; 150 − 30 − 12 − 8 + 5 flips the wrong pair.

4. (A) Factor the common 14: 14 × 100 − 14 × 3 = 14 × (100 − 3) = 14 × 97. Writing 14 × 97 − 3 subtracts the 3 once more after it was already inside the bracket; 14 × 103 adds instead of subtracts; 1400 − 3 distributes only the first term and forgets to multiply the 3 by 14.

5. (A) Brackets first: 2 + 4 × 2 = 2 + 8 = 10, and 11 − 2 × 4 = 11 − 8 = 3. The two divisions run left to right: 120 ÷ 10 = 12, then 12 ÷ 2 = 6; and 3 × 3 = 9; finally 6 + 9 = 15. Keeping only the bracket product 3 × 3 and dropping the 6 gives 9. Stopping after 120 ÷ 10 = 12 and never halving gives 12. Grouping the divisions as 120 ÷ (10 ÷ 2) = 24, which then plus 9 — or instead 120 ÷ 10 ÷ 2 = 6 plus a mis-multiplied 3 × (11 − 2) = 27 — gives 33.

6. (D) Each × or ÷ chain collapses to one signed term: 4 × 5 = 20 carries the minus in front, giving −20; 36 ÷ 9 = 4 carries its leading plus, giving +4; the standalone 90 and −7 stay. So the terms are 90, −20, 4, −7. Listing 90, −4, 5, 36, 9, −7 splits the products into separate numbers; making the +4 negative ignores its plus sign; making all positive ignores the operation signs.

7. (C) Going from 73 − 28 to 75 − 31 the first number rises by 2 but the amount subtracted rises by 3, a net change of 2 − 3 = −1, so 75 − 31 is one less. Hence 73 − 28 is greater. Calling them equal ignores that the subtrahend grew more than the minuend; choosing 75 − 31 assumes the bigger first number always wins.

8. (C) Subtracting 24 and then 16 removes 24 + 16 = 40 in total, so 100 − 24 − 16 = 100 − (24 + 16). Writing 100 − (24 − 16) removes only 8; (100 − 24) + 16 adds the 16 back; 100 + (24 − 16) adds instead of subtracts.

9. (A) Innermost: 2 × 3 = 6, then 8 + 6 = 14, then 20 − 14 = 6. Now 5 × 6 = 30, and 30 − 4 = 26. Forgetting to subtract the trailing 4 gives 30. Adding the 4 instead of subtracting it gives 34. Computing the inner 8 + 2 × 3 as 8 + 2 = 10 (skipping the × 3) gives 20 − 10 = 10, so 5 × 10 − 4 = 46.

10. (C) The common factor 6 pulls out: 6 × 48 + 6 × 52 = 6 × (48 + 52) = 6 × 100 = 600. Getting 100 stops at 48 + 52 and forgets the × 6; 300 multiplies only one of the two terms by 6; 6000 mis-places the zero.

11. (C) Reading the expression as signed terms 36, −19, +14, −6, the positives sum to 36 + 14 = 50 and the negatives to 19 + 6 = 25, giving 50 − 25 = 25; this regrouping is allowed because addition is commutative and subtraction is adding a negative. The choice that adds all four as positives (75) drops the minus signs; (36 + 19) − (14 + 6) wrongly moves +14 into the subtracted group; claiming order is fixed denies commutativity.

12. (C) The minus before the bracket flips all three inner signs: −(25 − 10 − 5) = −25 + 10 + 5, so the value is 80 − 25 + 10 + 5 = 70. Leaving every sign unchanged gives 40; flipping only two of the three signs gives 60; flipping the leading 25 the wrong way gives 90.

13. (B) Write both around 28×40: $27 \times 41 = (28 - 1)(40 + 1) = 28 \times 40 + 28 - 40 - 1 = 28 \times 40 - 13$, which is 13 less than 28×40. So 28×40 is greater. Calling them equal assumes shifting 1 from each factor cancels; choosing 27×41 trusts the larger-looking second factor.

14. (A) Brackets first: $3 + 3 \times 2 = 3 + 6 = 9$ and $9 - 2 \times 3 = 9 - 6 = 3$. Then left to right among $\times$ and $\div$: $72 \div 9 = 8$, $8 \times 4 = 32$, and $2 \times 3 = 6$; finally $32 - 6 = 26$. Treating $\times 4$ as part of the divisor, $72 \div (9 \times 4) = 2$ and dropping the rest leaves 2. Computing the second bracket as $9 - 2 = 7$ (skipping the $\times 3$) gives $32 - 2 \times 7 = 18$. Adding the last product instead of subtracting gives $32 + 6 = 38$.

15. (C) Since $101 = 100 + 1$, $23 \times 101 = 23 \times 100 + 23 \times 1 = 2300 + 23 = 2323$. Writing $23 \times 100 + 1$ multiplies only the 100 by 23 and forgets to multiply the 1; doubling to $23 \times 100 + 23 \times 100$ splits 101 as $100 + 100$; subtracting 23 treats 101 as $100 - 1$.

16. (D) Adult cost is 4×120, child cost is 3×80, and the single coupon subtracts 50 once: $4 \times 120 + 3 \times 80 - 50$. Reading it as $4 + 120 \times 3 + 80 - 50$ adds the counts to the prices wrongly; $(4 + 3) \times (120 + 80)$ charges every person both prices; 50×7 deducts the coupon once per ticket.

17. (B) Work outward removing brackets with sign flips: $10 - 3 = 7$, then $40 - 7 = 33$, then $100 - 33 = 67$; equivalently the nested minus signs flip to $100 - 40 + 10 - 3 = 67$. Keeping all signs negative gives 47; flipping the 3 to plus gives 73; flipping the 10 the wrong way gives 53.

18. (B) From the first sum to the second, one part rises by 3 ($248 \rightarrow 251$) while the other falls by 2 ($176 \rightarrow 174$), a net change of $+3 - 2 = +1$, so $251 + 174$ is one larger. Calling them equal assumes the $+3$ and -2 cancel; choosing $248 + 176$ misreads which side gained more.

19. (D) Inner brackets: $7 - 3 = 4$ so $2 \times 4 = 8$ and $4 + 8 = 12$, giving $3 \times 12 = 36$; and $6 - 2 \times 2 = 6 - 4 = 2$ giving $5 \times 2 = 10$. Then $36 - 10 = 26$. Adding the two products gives 46; subtracting 36 from a smaller piece gives 6; stopping at $3 \times 12 = 36$ forgets the second product entirely.

20. (D) Distributing the 9 over the difference multiplies both inner terms: $9 \times (20 - 4) = 9 \times 20 - 9 \times 4$. The form $9 \times 20 - 4$ multiplies only the 20 by 9; $(9 - 20) \times 4$ versus $9 - 20 \times 4$ changes which factor the bracket binds; $9 \times 20 + 9 \times 4$ adds instead of subtracts.

21. (D) The signed terms are 56, -20, $+18$, -2. Adding the positives $56 + 18 = 74$ and the negatives $20 + 2 = 22$ gives $74 - 22 = 52$. Dropping the $+18$ term gives $56 - 20 - 2 = 34$; making the -20 positive gives $56 + 20 + 18 - 2 = 92$; reading the last term as $+2$ gives $56 - 20 + 18 + 2 = 56$.

22. (B) Three successive subtractions remove the total 8 + 9 + 5 = 22, so the expression is 60 − (8 + 9 + 5). Writing 60 − (8 − 9 − 5) removes only 8 − 14 = −6; 60 − (8 + 9 − 5) adds the 5 back; the last option both splits and re-adds a 5.

23. (B) Since 34 × 36 = (35 − 1)(35 + 1) = 35 × 35 − 1, the product 34 × 36 is exactly one less than 35 × 35, so 35 × 35 is greater. Calling them equal assumes spreading the factors by 1 keeps the product; choosing 34 × 36 trusts that a larger factor (36) always wins.

24. (A) Innermost: 8 − 6 = 2, then 12 ÷ 2 = 6, then 144 ÷ 6 = 24; finally 24 + 2 × 3 = 24 + 6 = 30. Ignoring the inner bracket and dividing left to right, 144 ÷ 12 ÷ 2 = 6, then 6 + 6 = 12. Forgetting the trailing + 2 × 3 leaves the bare 24. Adding 2 before multiplying by 3, (24 + 2) × 3, gives 78.

25. (D) Total bottles are 9 crates of 6 each, 9 × 6 = 54, and 4 break, so 9 × 6 − 4 = 50. The form 9 × (6 − 4) wrongly removes 4 bottles from every crate; (9 − 4) × 6 removes whole crates instead of bottles; 9 + 6 − 4 adds crates to bottles.

26. (B) Since 93 = 100 − 7, 8 × 93 = 8 × 100 − 8 × 7. Writing 8 × 100 − 7 subtracts only 7 instead of 8 × 7; 8 × 90 + 3 splits 93 as 90 + 3 but forgets to multiply the 3 by 8; 8 × 100 + 8 × 7 treats 93 as 100 + 7.

27. (B) A plus before a bracket keeps the inner signs (+30 − 10); a minus before a bracket flips them (−8 + 3). So the result is 50 + 30 − 10 − 8 + 3. Leaving the second bracket's −3 unflipped gives … − 8 − 3; flipping the first (plus) bracket's signs gives 50 − 30 + 10 …; flipping the −10 to +10 leaves 50 + 30 + 10 …

28. (D) Collapse each × or ÷ chain to a signed term: 48 ÷ 6 = 8 (so −8), 5 × 4 = 20 (so +20), 12 ÷ 4 = 3 (so −3). Then 100 − 8 + 20 − 3 = 109. Making the +20 term negative gives 100 − 8 − 20 − 3 = 69; making the −8 positive gives 100 + 8 + 20 − 3 = 125; making the −3 positive gives 100 − 8 + 20 + 3 = 115.

29. (B) Because 99 = 100 − 1, the distributive idea gives 47 × 100 − 47 × 1 = 47 × 99, so both are identical. Choosing 47 × 100 − 47 as larger is misled by the big 47 × 100; choosing 47 × 99 forgets the subtraction undoes the extra group of 47.

30. (C) Work from the deepest bracket: 7 + 1 = 8, then 2 × 8 = 16, then 30 − 16 = 14, then 14 ÷ 2 = 7. Now 4 × 7 = 28, and 28 + 5 = 33. Skipping the ÷ 2 gives 4 × 14 + 5 = 61. Stopping at 4 × 7 = 28 and dropping the + 5 leaves 28. Subtracting the 5 instead of adding it gives 28 − 5 = 23.

31. (A) Each hour nets 90 − 40 = 50 and there are 7 hours, so 7 × (90 − 40) = 350; this also equals 7 × 90 − 7 × 40. The form 7 × 90 − 40 subtracts the spending for only one hour; 7 × 90 − 40 × 1 does the same; multiplying the whole bracket by 7 again over-counts the shift.

32. (B) Since 35 = 30 + 5, 18 × 35 = 18 × 30 + 18 × 5. Writing 18 × 30 + 5 forgets to multiply the 5 by 18; 18 × 30 × 5 multiplies the parts instead of adding them; the split is certainly possible by distribution.

33. (C) The minus flips every sign inside the bracket: −(15 + 6 − 4) = −15 − 6 + 4; the trailing + 5 is outside and unchanged. So it is 90 − 15 − 6 + 4 + 5. Flipping only the first sign gives 90 − 15 + 6 − 4; leaving the −4 unflipped gives ... − 6 − 4; flipping the +6 to keep + and +4 also misreads the inner signs.

34. (B) Evaluate each: (6 + 2) × (5 − 1) = 8 × 4 = 32; 6 + 2 × 5 − 1 = 6 + 10 − 1 = 15; 6 × 2 + 5 × 1 = 12 + 5 = 17; 6 × 2 − 5 + 1 = 12 − 5 + 1 = 8. The largest is 32. The trap is reading the bracketed product as if order of operations made it small, or computing 6 + 2 × 5 − 1 left-to-right as (6 + 2) × 5 − 1 = 39.

35. (C) From the first to the second, the minuend drops by 3 (502 → 499) and the subtrahend drops by 4 (318 → 314); a smaller subtraction of 4 outweighs the smaller start of 3, net +1, so 499 − 314 is larger by 1. Calling them equal assumes the two drops cancel; choosing 502 − 318 trusts the bigger first number.

36. (B) Innermost: 2 × 4 = 8, then 12 + 8 = 20, then 40 − 20 = 20; so 3 × 20 = 60 and 250 − 60 − 8 = 182. Failing to flip the inner bracket sign (writing 40 − 12 + 2 × 4 = 36, so 3 × 36 = 108) gives 250 − 108 − 8 = 134. Forgetting the trailing − 8 gives 190. Dropping the whole product 3 × (...) leaves 250 − 8 = 242.

37. (D) Each bus holds 32 + 2 = 34 people and there are 5 buses, so 5 × (32 + 2) = 170; equivalently 5 × 32 + 5 × 2. The form 5 × 32 + 2 counts the teachers on just one bus; 5 + 32 × 2 mixes the counts; (5 + 32) × 2 doubles a meaningless sum.

38. (C) The common factor 15 pulls out: 15 × 7 + 15 × 13 = 15 × (7 + 13) = 15 × 20. Adding an extra 15 (15 × 20 + 15) double-counts; 15 × 91 multiplies the 7 and 13 instead of adding them; 15 + 7 + 13 drops the multiplications entirely.

39. (D) Removing a bracket after a minus flips each inner sign: −(30 + 25) = −30 − 25, so 120 − (30 + 25) = 120 − 30 − 25. The version with − 30 + 25 fails to flip the +25; 120 − (30 − 25) should become 120 − 30 + 25, not − 25; the last option fails to flip the 30.

40. (D) Pairing 19 + 81 = 100 and 36 + 64 = 100 makes two round hundreds totalling 200; this regrouping is allowed because addition is commutative and associative. The pairing (19 + 36) + (81 + 64) gives 55 + 145 — correct total but no round pair; (19 + 64) + (36 + 81) gives 83 + 117 — again no round pair; addition can always be regrouped.

41. (A) Order: 6 ÷ 3 = 2 so the bracket is 5 + 2 = 7; the × and ÷ run left to right giving 96 ÷ 8 = 12, 12 × 3 = 36, and 2 × 7 = 14. Then 36 − 14 + 4 = 26. Subtracting the trailing 4 instead of adding it gives 36 − 14 − 4 = 18. Adding the bracket product instead of subtracting gives 36 + 14 + 4 = 54. Grouping the first chain as 96 ÷ (8 × 3) = 4 gives 4 − 14 + 4 = −6.

42. (C) Rewrite each as a count of 25s: 25 × 41 is 41 groups; 25 × 42 − 25 × 2 = 25 × 40 (40 groups); 25 × 38 + 25 × 2 = 25 × 40 (40 groups); 25 × 39 + 25 = 25 × 40 (40 groups). So 25 × 41 alone has 41 groups and is greatest, while the other three each reduce to 25 × 40. Picking 25 × 42 − 25 × 2 is the trap of overrating the leading 25 × 42 and undercounting the subtraction.

43. (B) Factor the first two terms: 17 × 6 + 17 × 4 = 17 × (6 + 4) = 17 × 10 = 170, then subtract the lone 17 to get 153. Stopping at 170 forgets the final − 17; adding the 17 gives 187; subtracting 17 twice gives 136.

44. (A) Every inner sign flips after the minus: −(50 − 30 + 20 − 10) = −50 + 30 − 20 + 10. So the expansion is 200 − 50 + 30 − 20 + 10. Keeping the original inner signs gives 200 − 50 − 30 + 20 − 10; flipping only some terms gives 200 − 50 + 30 + 20 − 10; flipping the leading 50 wrongly gives 200 + 50 − 30 ...

45. (B) Three drains of 80 remove 3 × 80 = 240, and 150 is added: 500 − 3 × 80 + 150 = 410. The form 500 − 3 + 80 + 150 treats the count and litres as separate additions; 500 − 3 × (80 + 150) wrongly drains the refill three times; (500 − 3) × 80 multiplies the wrong quantity.

46. (D) Within a × and ÷ chain we work left to right: 36 ÷ 3 = 12, then 12 × 2 = 24; the minus in front makes the term −24. Doing 3 × 2 = 6 first (wrongly prioritising ×) then 36 ÷ 6 = 6 gives −6; stopping at 36 ÷ 3 = 12 gives −12; dropping the leading minus gives +24.

47. (A) Brackets first: 84 + 36 = 120, 2 + 4 × 2 = 10, and 10 − 2 × 4 = 2. Then 120 ÷ 10 = 12 and 3 × 2 = 6; finally 12 − 6 = 6. Adding the two products instead of subtracting gives 12 + 6 = 18. Computing the second bracket as 10 − 2 = 8 (skipping the × 4) gives 3 × 8 = 24, so 12 − 24 = −12. Dividing only by the 2 (reading the divisor as just 2) gives 120 ÷ 2 = 60, so 60 − 6 = 54.

48. (A) In 3 × (20 + 4) the 3 multiplies both 20 and 4, giving an extra 3 × 4 = 12 beyond 3 × 20; in 3 × 20 + 4 the 4 is added only once. So 3 × (20 + 4) is larger by 12 − 4 = 8. Calling them equal is the half-distribution error of multiplying only the first inner term; the two are clearly comparable, so 'cannot be decided' is wrong.

49. (D) Each carton has 5 + 3 = 8 pens, so 12 cartons hold 12 × (5 + 3); distributing gives 12 × 5 + 12 × 3 = 96. The form 12 × 5 + 3 multiplies only the red pens by 12; 12 × 5 + 3 is not equal to 12 × (5 + 3); and 12 × 5 × 3 multiplies the colours together rather than adding them.

50. (C) Innermost: 12 − 4 = 8, then inside the square bracket 30 − 8 − 6 = 16, and finally 64 − 16 = 48. Treating the square bracket as 30 − 8 + 6 = 28 (mis-flipping the −6) gives 64

− 28 = 36... ; the value 32 comes from 64 − 32 using 30 − 8 + 6 + 4-style slips; 40 comes from 30 − 4 − 6 inside (dropping the 12); 56 comes from 64 − 8 keeping only the inner difference.

51. (D) First form the pouches: 84 ÷ 7 = 12, then give away 5, leaving 84 ÷ 7 − 5 = 7. The form 84 ÷ (7 − 5) divides by 2 pencils per pouch; (84 − 5) ÷ 7 removes 5 pencils instead of 5 pouches; 84 ÷ 7 ÷ 5 divides the pouches among 5.

52. (A) Each product or quotient becomes one signed term: 4 × 3 = 12 with its leading minus gives −12; 18 ÷ 2 = 9 with its leading plus gives +9; the standalone 60 and −5 remain. So the terms are 60, −12, 9, −5. Splitting into single numbers gives the six-number list; making the +9 negative drops its plus sign; making everything positive ignores all operation signs.

53. (D) Inner pieces: 4 + 2 = 6 so 3 × 6 = 18; and 2 × 3 = 6 so 10 − 6 = 4. The big bracket is 18 − 4 = 14, then 2 × 14 = 28, and 5 + 28 = 33. Failing to flip the second sub-bracket (using 18 + 4 = 22) gives 5 + 2 × 22 = 49. Forgetting the outer + 5 leaves 2 × 14 = 28. Reading 3 × (4 + 2) as 3 × 4 + 2 = 14 gives 5 + 2 × (14 − 4) = 25.

54. (B) The second equals 64 × (100 − 3) = 64 × 97, which is one group of 64 fewer than 64 × 98. So 64 × 98 is greater by 64. Calling them equal mistakes the subtraction of three 64s for two; choosing the second is misled by the large 64 × 100.

55. (D) Since 36 = 40 − 4, 25 × 36 = 25 × 40 − 25 × 4 = 1000 − 100 = 900. Writing 25 × 40 − 4 subtracts only 4 rather than 25 × 4; adding 25 × 4 treats 36 as 44; the arithmetic 1000 − 100 = 800 is a subtraction slip.

56. (A) Without brackets 20 − 6 + 2 = 16; to reach 12 we must subtract both the 6 and the 2, i.e. 20 − (6 + 2) = 20 − 8 = 12. The grouping (20 − 6) + 2 gives 16; 20 − (6 − 2) gives 16; bracketing the whole thing changes nothing, still 16.

57. (C) Each shirt costs 250 − 30 = 220 and there are 8, so 8 × (250 − 30) = 1760; equivalently 8 × 250 − 8 × 30. The form 8 × 250 − 30 applies the discount once for all eight; 8 × 250 − 30 × 1 does the same; multiplying the bracket by 8 again double-counts the order.

58. (C) Collapse each chain: 144 ÷ 12 = 12, 7 × 3 = 21, 50 ÷ 5 = 10, 2 × 2 = 4, with signs +12, +21, −10, −4. Sum: 12 + 21 − 10 − 4 = 19. Treating the −4 as +4 gives 12 + 21 − 10 + 4 = 27; treating the −10 as +10 gives 12 + 21 + 10 − 4 = 39; treating both subtracted terms as added gives 12 + 21 + 10 + 4 = 47.

59. (A) Evaluate the original: 40 + 20 = 60, 60 ÷ 6 = 10, 3 × 4 = 12, total 22; so 60 ÷ 6 + 12 matches. Writing 40 + 20 ÷ 6 + 12 drops the bracket and divides only the 20; (40 + 20) ÷ (6 + 3) × 4 absorbs 3 × 4 into the divisor; 60 ÷ (6 + 3 × 4) divides 60 by the whole tail.

60. (C) Since 99 = 100 – 1, subtracting 99 is the same as subtracting 100 then adding back 1: 200 – 99 = 200 – 100 + 1 = 101. The form 200 – 100 – 1 subtracts 101; 200 + 100 – 1 adds instead of subtracting; 200 – 90 – 9 – 1 subtracts 100 in total, giving 100, not 101.